

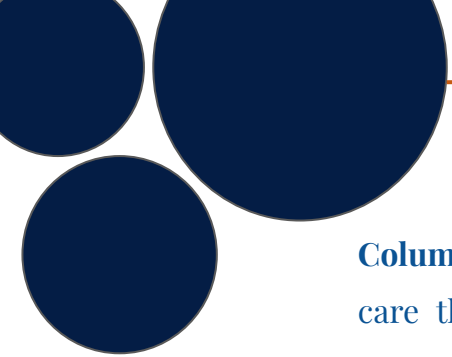


# COLUMBIA ASIA HOSPITAL ANALYSIS

Healthcare analytics &  
Power BI project Dashboard

Prepared By: Sanika Dhawale





# ABOUT US

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**Columbia Asia Hospital** is a modern healthcare organization focused on delivering quality patient care through efficient medical services, experienced healthcare professionals, and data-driven operational excellence. With growing patient volumes and increasing healthcare demands, the hospital aims to leverage analytics and business intelligence to improve decision-making, optimize resource allocation, and enhance overall patient experience.

**The hospital serves a diverse patient population across multiple departments, emphasizing:**

- Patient satisfaction
  - Efficient healthcare delivery
  - Reduced waiting time
  - Advanced medical support
  - Continuous operational improvement
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Strategy  
suggestions for  
patient discounts

Assess the  
hospital's revenue  
generation

Insights about  
suitable  
departments  
for new hires

## PROBLEM STATEMENT

You have been hired as a consultant data analyst by Columbia Asia Hospital. Your task is to perform data analysis and come up with a report in order to help the organization with the mentioned objectives.

# DATASET OVERVIEW

| patient_id  | patient_gender | patient_age | patient_sat_score | patient_first_initial | patient_last_name | patient_race | patient_admin_flag | patient_waittime | department_referral | Date    |
|-------------|----------------|-------------|-------------------|-----------------------|-------------------|--------------|--------------------|------------------|---------------------|---------|
| 603-14-9459 | Male           | 69          |                   | A                     | Milier            | White        | False              | 55               | General Practice    | 1       |
| 770-82-7913 | Male           | 66          |                   | N                     | Longworth         | White        | False              | 45               | General Practice    | 0       |
| 240-68-3060 | Male           | 66          |                   | I                     | Tootell           | White        | False              | 29               | General Practice    | 1       |
| 166-20-2794 | Male           | 68          |                   | U                     | Percy             | White        | True               | 44               | General Practice    | 28      |
| 525-36-7812 | Male           | 58          |                   | G                     | Thorley           | White        | False              | 14               | General Practice    | 20 Fe   |
| 177-05-5852 | Male           | 71          |                   | I                     | Delatour          | White        | False              | 25               | General Practice    | C       |
| 334-31-4472 | Male           | 61          |                   | C                     | Spracklin         | White        | False              | 30               | General Practice    | 0       |
| 500-12-4114 | Male           | 56          |                   | E                     | Connal            | White        | True               | 15               | General Practice    | C       |
| 413-62-4921 | Male           | 57          |                   | R                     | Walsham           | White        | True               | 56               | General Practice    | 1       |
| 818-72-0858 | Male           | 65          |                   | L                     | Mackilpatrick     | White        | True               | 15               | General Practice    | 19      |
| 827-11-0047 | Male           | 69          |                   | Y                     | Hatherleigh       | White        | True               | 46               | General Practice    | 10 Sept |
| 664-09-7964 | Male           | 63          |                   | N                     | O'Dulchonta       | White        | True               | 13               | General Practice    | 21 C    |
| 844-55-2304 | Male           | 74          |                   | R                     | Paroni            | White        | False              | 22               | General Practice    | 09      |
| 812-72-2025 | Male           | 65          |                   | E                     | Whitwam           | White        | False              | 54               | General Practice    | 3       |
| 774-37-2696 | Male           | 73          |                   | Y                     | Mauchlen          | White        | True               | 53               | General Practice    | 0       |
| 468-02-7614 | Male           | 56          |                   | P                     | Dmisek            | White        | True               | 32               | General Practice    | 31 Dec  |
| 177-54-8118 | Male           | 73          |                   | J                     | Errigo            | White        | False              | 35               | General Practice    | .       |
| 281-13-9122 | Male           | 74          |                   | G                     | Goodenough        | White        | False              | 59               | General Practice    | 22 C    |
| 808-32-5113 | Male           | 57          |                   | C                     | Ginni             | White        | True               | 27               | General Practice    | 06 C    |
| 201-10-9557 | Male           | 69          |                   | F                     | Relfe             | White        | False              | 36               | General Practice    | 1       |
| 177-26-9806 | Male           | 65          |                   | X                     | MacGibbon         | White        | True               | 16               | General Practice    | 09 C    |
| 662-33-7487 | Male           | 56          |                   | M                     | Fancy             | White        | True               | 47               | General Practice    | 19 C    |
| 727-81-2340 | Male           | 68          |                   | P                     | Capron            | White        | True               | 20               | General Practice    | 17      |
| 804-99-9395 | Male           | 67          |                   | R                     | Cragg             | White        | False              | 16               | General Practice    | 17 Sept |
| 400-06-0553 | Male           | 66          |                   | V                     | Mourant           | White        | True               | 22               | General Practice    | 12 Nov  |
| 820-83-9876 | Male           | 62          |                   | A                     | Castelyn          | White        | False              | 33               | General Practice    | 10      |
| 155-22-9624 | Male           | 69          |                   | O                     | Snodden           | White        | True               | 47               | General Practice    |         |

# METHODOLOGY

**Data Collection & Preparation** — Collected hospital operational data and performed data cleaning, formatting, and validation for accurate analysis.

**SQL & Data Analysis** — Used SQL queries and analytical techniques to identify trends, department performance, and patient behavior insights.

**Power BI Dashboard Development** — Created interactive dashboards and KPI visualizations to monitor hospital performance and operational metrics.

**Business Insights & Recommendations** — Generated data-driven insights and strategic recommendations to improve efficiency, patient satisfaction, and revenue growth.

# POWER BI DASHBOARD OVERVIEW

An interactive 3-page Power BI dashboard giving hospital management real-time visibility into visits, revenue, satisfaction, and wait times enabling doctor- and department-level analysis for data-driven decisions.

## *Dashboard Components*



**Main Dashboard** – Hospital-wide performance monitoring



**Doctors Dashboard** – Doctor productivity & revenue analysis



**Patients Dashboard** – Patient demographics & visit insights

## KEY FEATURES

- Interactive slicers & drill-down analysis
- Real-time KPI monitoring
- Revenue, satisfaction & waiting-time tracking
- Actionable insights for management

# COLOMBIA HOSPITAL MANAGEMENT ANALYSIS



Date

01-04-2019 30-10-2020

Departments

- ☐ Cardiology
- ☐ Gastroenterology
- ☐ General Practice
- ☐ Neurology
- ☐ Orthopedics
- ☐ Physiotherapy
- ☐ Renal

Total Revenue

509M

Avg Wait Time

35.26

Avg SAT Score

5.47

Total Visits

9216

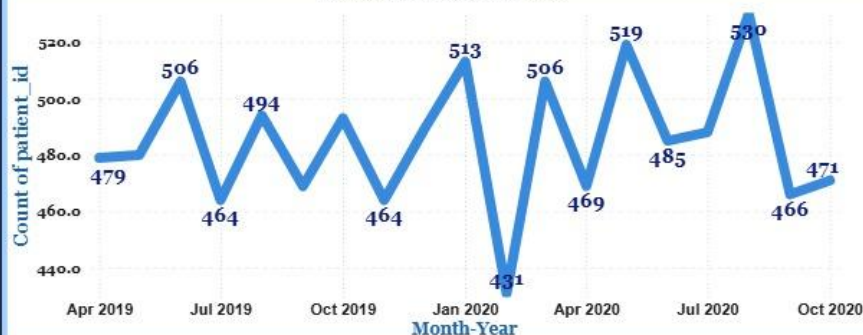
Total Departments

7

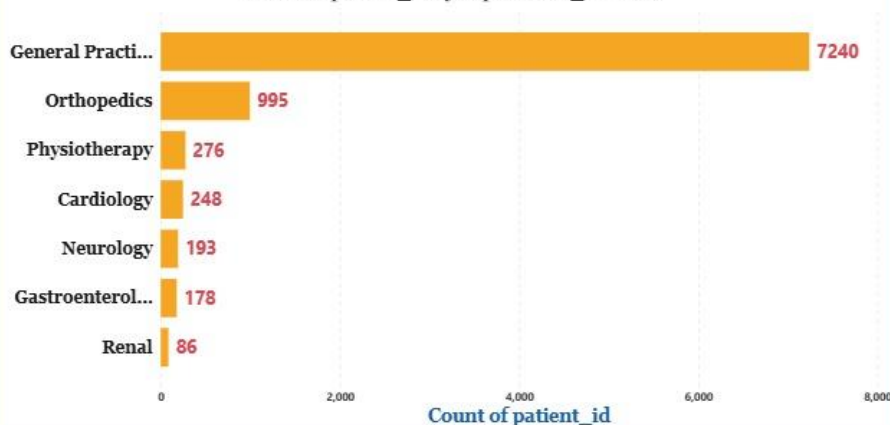
Patient Count per Hour



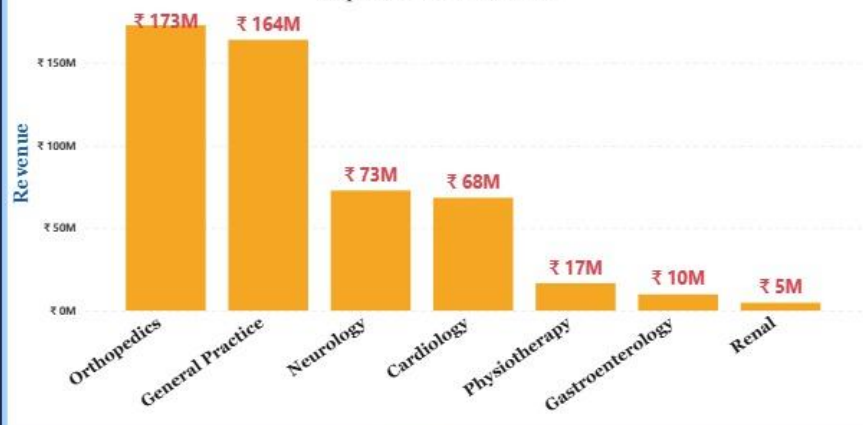
Patient Volume Over Time



Count of patient\_id by department\_referral



Department wise Revenue



# COLOMBIA HOSPITAL — PATIENT ANALYSIS

Avg Wait Time  
**35.26**

Total Visits  
**9216**

Avg SAT Score  
**5.47**

Avg Age  
**39.86**

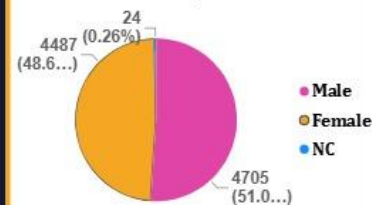
Age Group

0-18

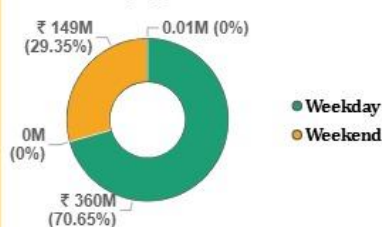
19-35

36-55

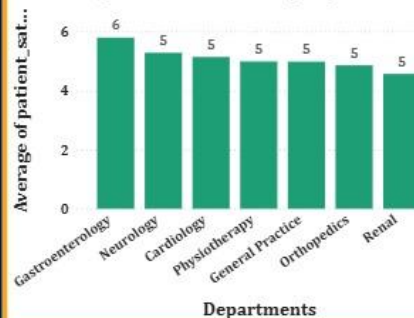
Count by Gender



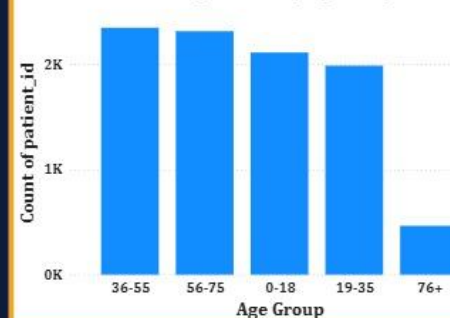
Day Type wise Revenue



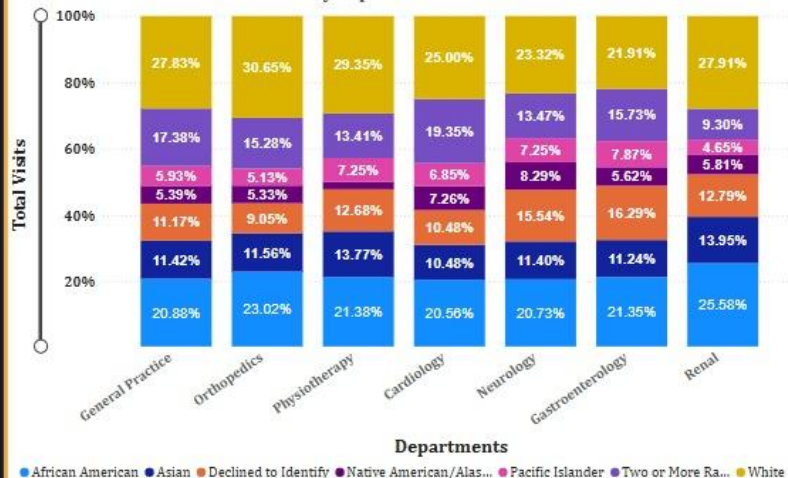
Average Patient's SAT Score by Departments



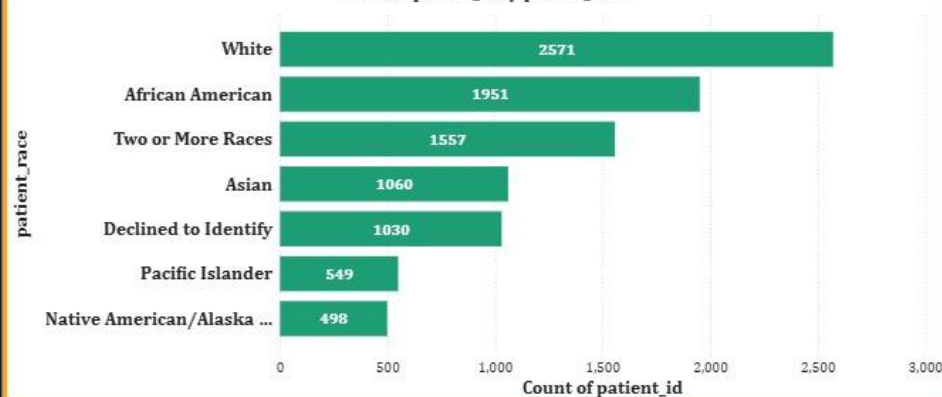
Count of patient\_id by Age Group



Total Visits by Departments and Patient's Race



Count of patient\_id by patient\_race



# COLOMBIA HOSPITAL — DOCTORS ANALYSIS

Doctor's Count  
**22**

Avg Appointment Fee  
**580.21**

Total Appointments  
**9K**

Avg Revenue Per Doctor  
**23.15M**

Total Revenue  
**509M**

Doctor Name

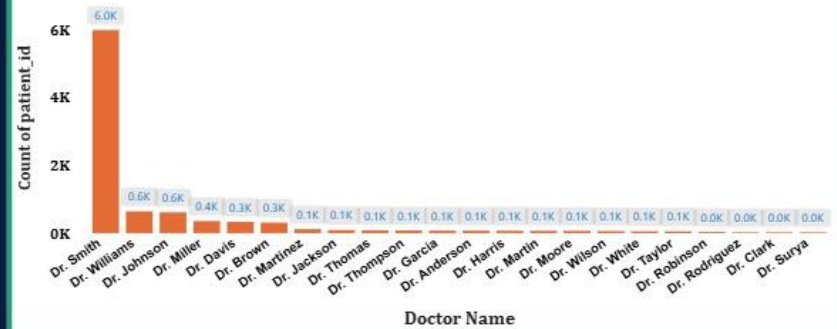
Enter Doctor Name

Department

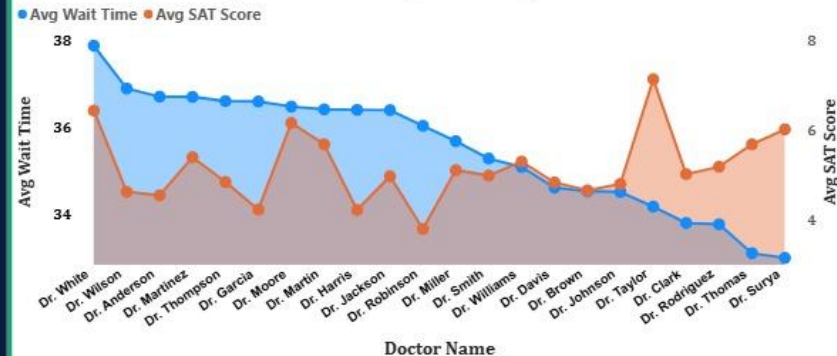
Enter Department

| Doctor        | Department       | Appointment Fees | Revenue        |
|---------------|------------------|------------------|----------------|
| Dr. Anderson  | Cardiology       | 1200             | ₹ 2,03,10,717  |
| Dr. Jackson   | Cardiology       | 1200             | ₹ 2,46,63,224  |
| Dr. Thomas    | Cardiology       | 1200             | ₹ 2,33,96,309  |
| Dr. Moore     | Gastroenterology | 700              | ₹ 35,92,727    |
| Dr. Surya     | Gastroenterology | 700              | ₹ 82,595       |
| Dr. Taylor    | Gastroenterology | 700              | ₹ 29,49,116    |
| Dr. Wilson    | Gastroenterology | 700              | ₹ 31,58,897    |
| Dr. Johnson   | General Practice | 500              | ₹ 1,39,32,626  |
| Dr. Smith     | General Practice | 500              | ₹ 13,56,79,687 |
| Dr. Williams  | General Practice | 500              | ₹ 1,44,58,503  |
| Dr. Harris    | Neurology        | 1500             | ₹ 2,66,82,124  |
| Dr. Martin    | Neurology        | 1500             | ₹ 2,60,58,420  |
| Dr. White     | Neurology        | 1500             | ₹ 2,00,55,208  |
| Dr. Brown     | Orthopedics      | 700              | ₹ 5,40,75,456  |
| Dr. Davis     | Orthopedics      | 700              | ₹ 5,74,13,306  |
| Dr. Miller    | Orthopedics      | 700              | ₹ 6,13,22,460  |
| Dr. Surya     | Orthopedics      | 700              | ₹ 1,28,551     |
| Dr. Garcia    | Physiotherapy    | 1000             | ₹ 45,35,780    |
| Dr. Martinez  | Physiotherapy    | 1000             | ₹ 70,61,544    |
| Dr. Surya     | Physiotherapy    | 1000             | ₹ 2,82,833     |
| Dr. Thompson  | Physiotherapy    | 1000             | ₹ 47,12,667    |
| Dr. Clark     | Renal            | 500              | ₹ 13,49,533    |
| Dr. Robinson  | Renal            | 500              | ₹ 19,70,689    |
| Dr. Rodriguez | Renal            | 500              | ₹ 14,36,145    |

Patient Count Per Doctor



Avg Wait Time and Avg SAT Score by Doctor Name



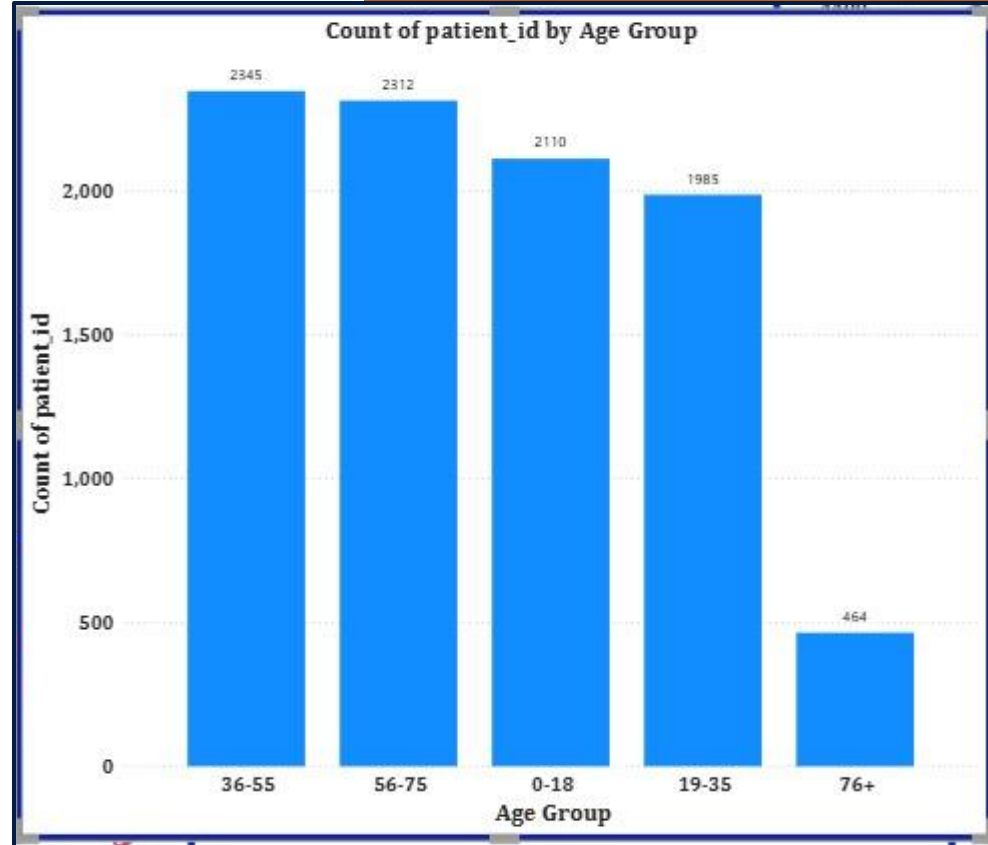
# PATIENTS AND AGE GROUP

## KEY FINDINGS :

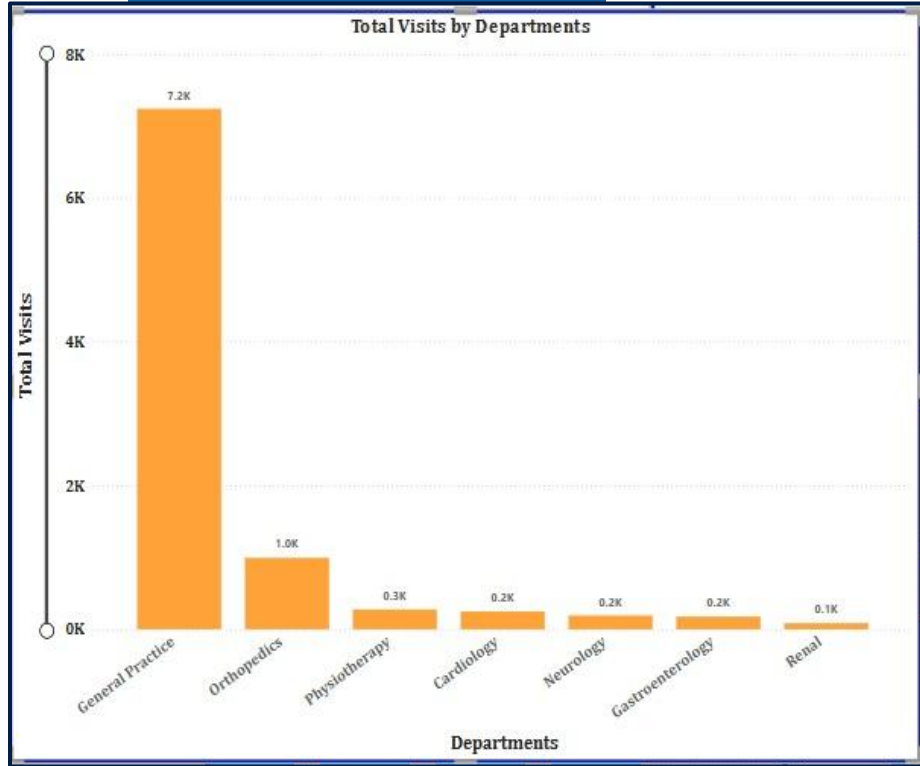
- 36–55 (2,345) and 56–75 (2,312) are the top two age groups, together ~50% of visits
- 0–18 (2,110) and 19–35 (1,985) each contribute roughly a fifth of visits
- 76+ is the smallest group at just 464 visits (5%)

## RECOMMENDATIONS :

Prioritize staffing for the 36–75 range while maintaining strong pediatric care and improving outreach to the underrepresented 76+ segment.



# PATIENTS VISIT & DEPARTMENTS



## KEY FINDINGS

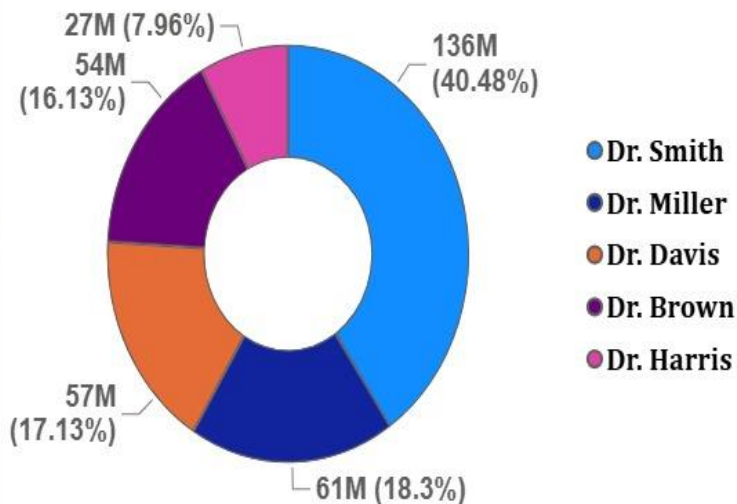
- General Practice dominates with 7.2K visits — over 75% of total volume
- Orthopedics is a distant second at 1.0K visits (~10%)
- All other departments (Physiotherapy, Cardiology, Neurology, Gastroenterology, Renal) each account for under 3% of visits

## RECOMMENDATIONS:

Ensure General Practice is adequately staffed to handle its outsized share of demand, and evaluate whether low-volume specialty departments need referral pathways, marketing, or scheduling adjustments to improve utilization.

# TOP 5 REVENUE GENERATING DOCTORS

Total Revenue by Doctor Name



## KEY FINDINGS

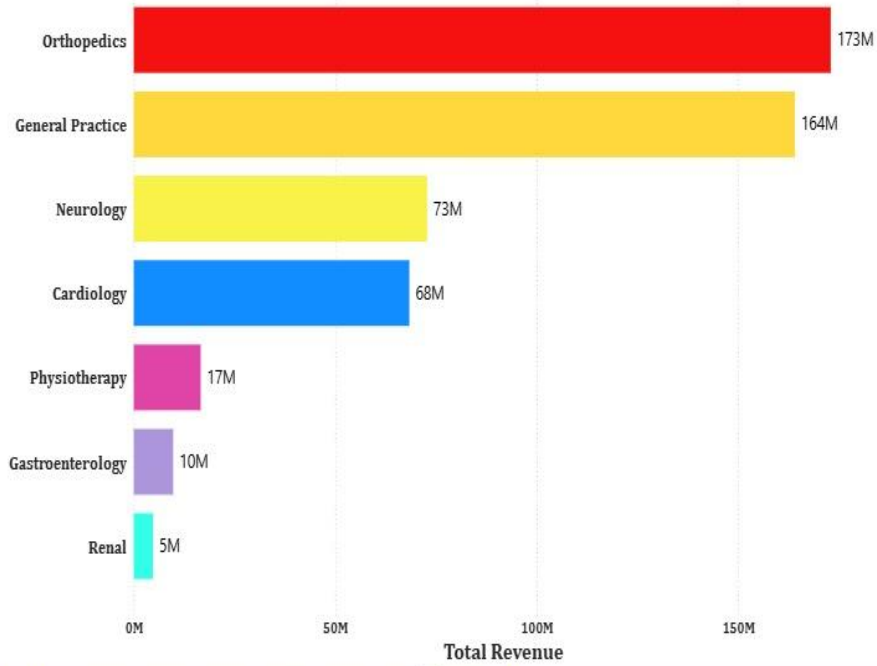
- Dr. Smith generates the most revenue at 136M (40.48%) — more than double any other doctor
- Dr. Miller and Dr. Davis follow at 61M (18.3%) and 57M (17.13%) respectively
- Dr. Brown contributes 54M (16.13%), while Dr. Harris trails at 27M (7.96%)
- Top 3 doctors (Smith, Miller, Davis) together account for ~76% of total doctor-driven revenue

## BUSINESS IMPACT:

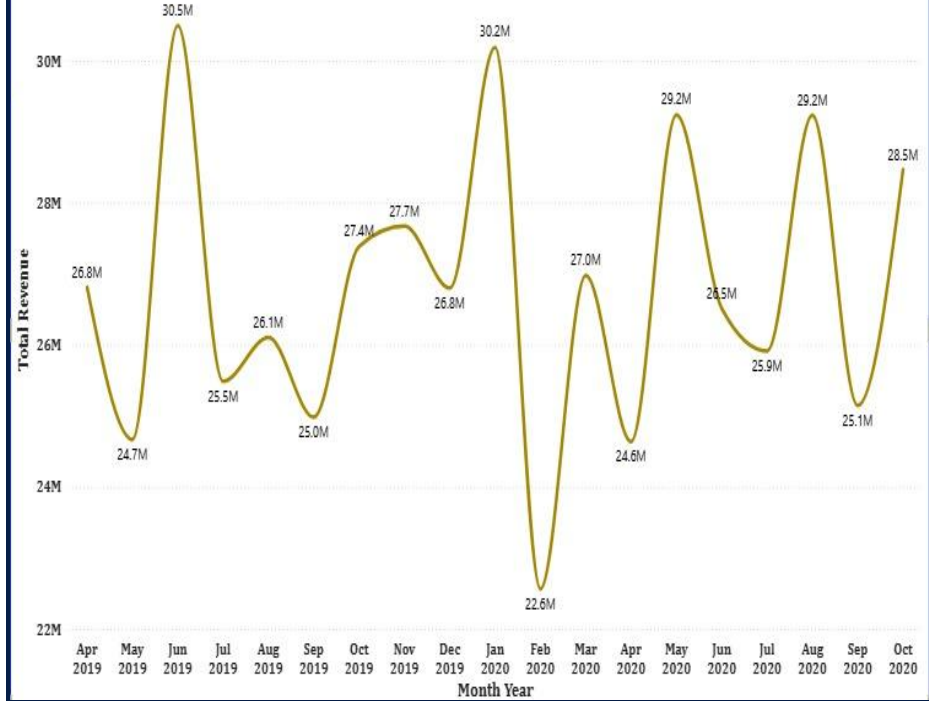
Dr. Smith's revenue concentration creates retention risk, while Dr. Harris's lower output signals room to rebalance patient load and boost overall revenue.

# PROFITABILITY ANALYSIS

Total Revenue by Department



Total Revenue by Month Year



# PROFITABILITY ANALYSIS

## PROFITABILITY INDICATORS

- ✓ Strong overall revenue base (\$509.31M)
- ✓ Multiple high-performing departments
- ✓ Consistent monthly revenue generation
- ⚠ Declining monthly revenue trend requires monitoring
- ⚠ Revenue concentration risk in two major departments

## CONCLUSION :

Based on revenue performance, the hospital appears financially healthy and likely profitable. However, the declining revenue trend and dependence on Orthopedics and General Practice highlight the need for revenue diversification and continued patient growth initiatives.

# DISCOUNT STRATEGY

## PROPOSED DISCOUNT STRATEGIES

-  **Tier-Based Discounts** – Offer 5–10% discounts based on patient bill value to encourage higher spending
-  **Department Bundles** – Create discounted service packages in high-demand departments to increase utilization
-  **Seasonal Promotions** – Introduce offers during low-visit periods to boost patient traffic
-  **New Patient Offers** – Provide first-visit discounts to attract and acquire new patients
-  **Loyalty Programs** – Reward long-term patients through memberships and exclusive healthcare benefits
-  **Community Discounts** – Offer targeted concessions for seniors, students, and special community groups

| PATIENT SEGMENT    | DISCOUNT |
|--------------------|----------|
| Wait Time > 30 min | 10%      |
| Frequent Visitors  | 15%      |
| Low Satisfaction   | 10%      |
| Senior Citizens    | 12%      |

# HIRING RECOMMENDATION



## Recommendations:

Hire 2 doctors in General Practice, which handles ~80% of visits and shows severe workload imbalance (Dr. Smith overloaded). Hire 1 doctor in Neurology, which has the highest wait time despite lower volume. Optionally, monitor Physiotherapy/Orthopedics for future hiring if wait times or satisfaction decline.

# RECOMMENDATIONS:

- **Prioritize staffing and resources** in Orthopedics & General Practice to support demand and revenue.
  - **Reduce waiting times** through optimized scheduling, staffing, and resource allocation.
  - **Monitor patient satisfaction regularly** and address service gaps proactively.
  - **Focus retention initiatives** on the high-volume **36–60 age group**.
  - **Replicate best practices** from high-performing doctors to improve overall doctor productivity.
  - **Use targeted promotions and packages** instead of blanket discounts to drive patient visits.
  - **Track revenue, visits, waiting time & satisfaction KPIs** regularly to identify declining trends early.
  - **Develop growth plans for lower-performing departments** while continuing to invest in high-revenue departments.
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*THANK YOU !*